Exploratory Data Analysis: Continuous University Funding Intensity and the Future Tax Take

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1 Overview, Continuous Treatment Design, and Holdout Protocol

This exploratory data analysis (EDA) audits the global fiscal and tertiary education panel (`data/processed/clean_tax_panel.csv`, spanning 1,736 country-waves across 217 sovereign countries from `1985-1989` to `2020-2024`) prior to estimating our continuous-treatment Multivariate Bayesian Causal Forest (`MV-BCF`) models. Our primary hypothesis is that higher national government expenditure on tertiary (university) education increases a country's future tax take at the $+5$-year and $+10$-year horizons, and that this fiscal return operates specifically through direct income, profit, and capital gains taxation rather than indirect consumption taxes.

To ensure leakage-free causal and out-of-sample evaluation, our Phase 2 and Phase 3 workflow enforces four design safeguards:

1. **Country-Blocked Holdout Split Locked Prior to EDA:** All 217 sovereign countries are partitioned via an income-group stratified 5-fold country-blocked split (`data/processed/splits/country_folds.csv`, `set.seed(20261103)`). Folds 1 to 4 form the **80% Training Set** (`98` countries / `301` country-waves at $+5$ years; `90` countries / `229` country-waves at $+10$ years), while Fold 5 forms the **20% Locked Holdout Test Set** (`23` unseen countries / `70` country-waves at $+5$ years; `22` unseen countries / `54` country-waves at $+10$ years). All income-group centring medians and all Phase 3 EDA diagnostics in Sections 3 to 7 of this report are computed strictly on the 80% Training Set.
2. **Dual Continuous University Funding Scalings ($Z_{i,t} \in \mathbb{R}$):** Rather than dichotomising university funding at an arbitrary median boundary, we retain the full continuous dose of public tertiary investment and evaluate two complementary scalings side by side:
   - **Scaling A (`Z_cont_10pp`, Additive Per-Student Intensity):** Winsorised public expenditure per tertiary student (`[1, 150]` % of GDP per person), centred on each World Bank income group's training-set median and divided by 10.0, so that $+1$ unit represents a **$+10$ percentage point increase in per-student funding (% of GDP per person)**.

- **Scaling B (`Z_cont_log25pct`, Multiplicative Log-Relative Intensity):** Natural log of public expenditure per tertiary student, centred on each World Bank income group's training-set log median and divided by $\log(1.25) \approx 0.2231$, so that $+1$ unit represents a $+25\%$ **relative increase in per-student funding**.
- **Unbundled Continuous Denominators:** We also construct continuous real 2021 PPP dollars per student (`Z_cont_real_ppp_log25pct`, removing GDP per person from the denominator), continuous public tertiary budget share of GDP (`Z_cont_budget_gdp_025pp`, removing student headcount from the denominator), and within-country 5-year per-student funding changes (`Z_cont_delta_10pp`).

3. **Random Forest Expected-Treatment Conditioning $\hat{z}(X_{i,t}) = \mathbb{E}[Z_{i,t} \mid X_{i,t}]$:** For continuous treatments, the binary propensity score $P(Z = 1 \mid X)$ is replaced by the pre-treatment expected-treatment score $\hat{z}(X_{i,t}) = \mathbb{E}[Z_{i,t} \mid X_{i,t}]$ estimated via a 500-tree Random Forest (`ranger`) across all 19 pre-treatment baseline covariates. Including $\hat{z}(X_{i,t})$ inside the prognostic forest $\mu(X_{i,t}, \hat{z}_{i,t})$ prevents regularisation-induced confounding by allowing the prognostic trees to partial out expected policy intensity before estimating the heterogeneous causal slope $\tau(X_{i,t}) \cdot Z_{i,t}$.
4. **Four Pure Disaggregated Tax Outcomes (Zero GDP Product Identities):**
   - Y_1: **Domestic Non-Resource Tax-to-GDP Change (Δ Tax/GDP, percentage points of GDP)**, paired with unadjusted World Bank WDI Δ Tax/GDP as a sensitivity check.
   - Y_2: **Direct Income, Profit, and Capital Gains Tax-to-GDP Change (Δ IncomeTax/GDP, percentage points of GDP)**.
   - Y_3: **Direct Income and Profit Share of Revenue Change (Δ IncomeTax/Rev, percentage points of revenue)**.
   - Y_4: **Indirect Goods and Services (Consumption / VAT) Share of Revenue Change (Δ GoodsTax/Rev, percentage points of revenue)**, which serves as a built-in specificity (negative-control) channel.

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2 Sample Coverage, Quality Screening, and Train/Holdout Split

Before constructing the estimation samples, our hygiene audit in `src/r/02_build_clean_tax_panel.R` verifies zero duplicate `(iso3, wave_index)` keys and screens out four physically impossible World Bank WDI tax-to-GDP entries (`<0.5%` or `>60.0%` of GDP: Sudan in `1995-1999` at `629.19%`, Timor-Leste in `2010-2014` at `109.99%`, and the United Arab Emirates in `2010-2014` at `0.34%` and `2015-2019` at `0.23%`) plus one UNESCO tertiary per-student unit-scale error (`<1.0%` of GDP per person: Madagascar in `2015-2019` at `0.0359%`, where a 10^{-3} decimal shift occurred relative to its adjacent `99.1%` and `151.4%` waves). Table 1 documents sample coverage, attrition, and the 80% / 20% country-blocked split.

Table 1: Sample coverage, quality screening, and stratified country-blocked train/holdout split (1985-1989 to 2020-2024). Shares in rows 6-7 and 9-10 denote percentages of the complete +5 y and +10 y panels.

Stage	Sample Definition	Country-Waves	Countries	Share (%)
1. Full Global Panel	217 sovereign countries × 8 waves (1985-2024)	1,736	217	100.0%
2. Valid Tax / GDP	Observed non-resource tax-to-GDP after anomaly screen	914	159	52.6%
3. 4-Channel Tax Vector	All 4 disaggregated tax outcomes observed at wave t	883	157	50.9%
4. Observed Continuous Z	Valid UNESCO per-student tertiary spend ($\geq 1\%$ GDP/pc)	511	166	29.4%
5. Full +5 y Sample	Complete 4-channel changes over $[t, t+1]$ + continuous Z	**371**	**121**	**21.4%**
6. +5 y Training (80%)	**Training countries (Folds 1 to 4) for EDA & fitting**	**301**	**98**	**81.1%**
7. +5 y Holdout (20%)	**Locked holdout test countries (Fold 5) for validation**	**70**	**23**	**18.9%**
8. Full +10 y Sample	Complete 4-channel changes over $[t, t+2]$ + continuous Z	**283**	**112**	**16.3%**
9. +10 y Training (80%)	**Training countries (Folds 1 to 4) for EDA & fitting**	**229**	**90**	**80.9%**
10. +10 y Holdout (20%)	**Locked holdout test countries (Fold 5) for validation**	**54**	**22**	**19.1%**

3 Unbundling the UNESCO Denominator Trap and Comparing Scalings A & B

Why is unbundling the tertiary spending denominator and comparing linear (Z_cont_10pp) versus log-relative (Z_cont_log25pct) scalings essential in cross-national work? Table 2 compares the continuous tertiary spending indicators and baseline fiscal structure across the four World Bank income groups within the 80% Training Set.

Table 2: Training-set tertiary spending distributions, continuous treatment standard deviations for Scaling A (Z_cont_10pp) and Scaling B (Z_cont_log25pct), and baseline fiscal structure by World Bank income group.

Income Group (Train Countries, Waves)	Median UNESCO Ratio (IQR)	SD(Z_{10pp}) (Scaling A)	SD($Z_{log25\%}$) (Scaling B)	**Real PPP $ / Student**	**Tertiary Budget (% GDP)**	Direct Income Tax Share (%)
Low income (17, 67)	**183.8%** (203.4%)	3.17	3.39	$3,953	0.47%	27.5%
Lower middle (33, 139)	68.9% (107.5%)	5.23	5.33	$3,162	0.65%	30.9%
Upper middle (44, 176)	27.6% (21.2%)	3.01	3.22	$3,631	0.66%	34.7%
High income (58, 278)	26.4% (13.4%)	**1.47**	**1.86**	**$10,906**	**0.94%**	**38.3%**

Three structural insights emerge from Table 2:

1. **The Cross-Income-Group Denominator Paradox:** In **Low income** training countries, median gross university enrolment is only 2.8%, so dividing a modest public university budget (0.47% of GDP) by a small elite student cohort and a low GDP per person yields an un-winsorised median UNESCO ratio of **183.8%** (IQR = 203.4%). Conversely, in **High income** training countries where 59.9% of the cohort attends university, the median UNESCO ratio is **26.4%** (IQR = 13.4%), even though high-income states spend nearly three times as much per student in real purchasing power (**$10,906** vs **$3,953**) and twice as much of their GDP on public universities (**0.94%** vs **0.47%**). Centring both Z_cont_10pp and Z_cont_log25pct on each country's own

training-set World Bank income-group median removes this cross-tier baseline inversion while preserving full continuous within-tier variation.

2. **Scale Heteroscedasticity in Scaling A vs Scaling B:** Under the additive linear scaling (`Z_cont_10pp`), a $+1$ unit change ($+10$ percentage points of GDP per person) represents a `38%` relative funding expansion in a typical High income country (median `26.4%`), but only a `5.4%` relative expansion in a typical Low income country (median `183.8%`). Furthermore, because over half of Low income observations exceed `150%`, winsorising linear `Z_cont_10pp` at `[1, 150]` clamps the upper half of Low income country-waves at the ceiling.
3. **Why Scaling B (`Z_cont_log25pct`) Stabilises Cross-Tier Leverage:** Defining `Z_cont_log25pct` on the natural log scale ($\log(\text{Spend}) - \text{Median}_{\text{inc}}(\log(\text{Spend}))$ divided by $\log(1.25)$) naturally compresses extreme right-tail ratios without artificial ceiling clamping and ensures that $+1$ unit corresponds to the same $\mathbf{+25\%}$ **proportional funding expansion** at every stage of economic development. We therefore fit and audit our Multivariate BCF models under both Scaling A and Scaling B in Phase 4 to test whether linear or proportional dose-response fits the data better.

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4 Training-Set Raw Continuous Slopes and Bivariate Dose-Response

Table 3 and Figure 1 present the raw, unadjusted continuous OLS slopes (per $+1$ unit of Z) and treatment-tertile means across all four fiscal outcomes in the 80% Training Set before any covariate or causal forest adjustment.

Table 3: Unadjusted training-set continuous linear slopes (percentage points per $+1$ unit of Z) and lowest vs highest treatment-tertile means across the four fiscal outcomes.

Scaling & Horizon	Summary Group	Waves (N)	Countries	Δ Non-Res Tax/GDP	Δ Income Tax/GDP	**Δ Income Tax Share**	Δ Goods Tax Share
Scaling A (+5 y)	Linear OLS Slope	301	98	−0.044	+0.007	+0.023	+0.039
Scaling A (+5 y)	Tertile 1 (Lowest Z)	101	48	+0.43	+0.32	+1.34	+0.68
Scaling A (+5 y)	Tertile 3 (Highest Z)	100	46	+0.23	**+0.40**	**+1.77**	+0.49
Scaling A (+10 y)	Linear OLS Slope	229	90	+0.022	+0.030	−0.063	+0.153
Scaling A (+10 y)	Tertile 1 (Lowest Z)	77	45	+0.52	+0.59	+3.00	+0.76
Scaling A (+10 y)	Tertile 3 (Highest Z)	76	39	**+0.87**	**+0.85**	**+3.20**	+0.77
Scaling B (+5 y)	Linear OLS Slope	301	98	−0.058	−0.004	+0.012	−0.068
Scaling B (+5 y)	Tertile 1 (Lowest Z)	101	48	+0.42	+0.31	+1.38	+0.52
Scaling B (+5 y)	Tertile 3 (Highest Z)	100	42	+0.16	**+0.36**	**+1.73**	+0.35
Scaling B (+10 y)	Linear OLS Slope	229	90	−0.010	−0.012	−0.185	−0.037
Scaling B (+10 y)	Tertile 1 (Lowest Z)	77	45	+0.54	+0.54	+2.62	+0.82
Scaling B (+10 y)	Tertile 3 (Highest Z)	76	40	**+0.79**	**+0.81**	**+3.20**	+0.64

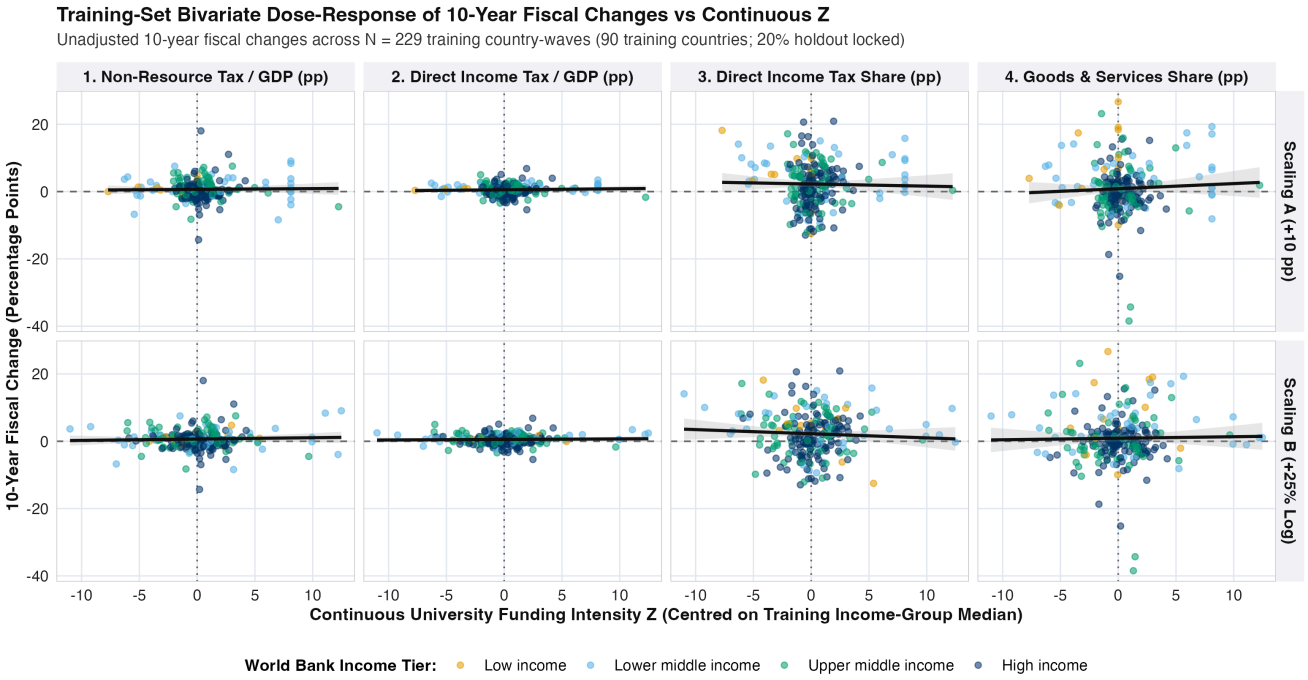


Figure 1: Training-set bivariate scatter and unadjusted linear trends of 10-year fiscal changes (percentage points) against continuous university funding intensity under Scaling A (top row) and Scaling B (bottom row) across N = 229 training country-waves (90 training countries).

Comparing the tertile means against the raw linear OLS slopes in Table 3 and Figure 1 reveals a critical lesson:

- **Why Unadjusted Linear Slopes Are Attenuated by Tail Outliers and Confounding:** Across the middle and upper tertiles of university funding, countries in the top tertile (`Tertile 3`) achieve consistently larger +10-year gains in total non-resource tax-to-GDP (`+0.87` vs `+0.52` pp under Scaling A; `+0.79` vs `+0.54` pp under Scaling B), direct income tax-to-GDP (`+0.85` vs `+0.59` pp; `+0.81` vs `+0.54` pp), and direct income tax share (`+3.20` vs `+3.00` pp; `+3.20` vs `+2.62` pp) than the bottom tertile. However, a naive unadjusted linear OLS line across all training observations (`Linear OLS Slope` near zero) is pulled flat by high-leverage lower-middle-income outliers with extreme baseline ratios (`|Z| > 6`) and strong baseline confounding (for example, higher baseline tax-to-GDP ratios naturally mean-revert downward, $r = -0.181$, while positively correlating with Z, $r = +0.291$).
- **Implications for Phase 4 Modelling:** This non-linear tail behaviour and baseline confounding directly motivate (1) conditioning on the Random Forest expected-treatment score $\hat{z}(X_{i,t})$ and baseline covariates $X_{i,t}$ in the prognostic forest $\mu(X_{i,t}, \hat{z}_{i,t})$, and (2) carefully auditing model residuals and tail leverage between Scaling A (`Z_cont_10pp`) and Scaling B (`Z_cont_log25pct`).

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5 Random Forest Expected-Treatment Fit Audit ($\hat{z}(X_{i,t}) = \mathbb{E}[Z_{i,t} \mid X_{i,t}]$)

To replace the binary propensity score in our continuous-treatment Multivariate BCF, we fit a non-parametric Random Forest (`ranger`, 500 trees, `mtry = 4`, `min.node.size = 5`) predicting continuous $Z_{i,t}$ from the 19 pre-treatment covariates $X_{i,t}$ on the 80% Training Set ($N = 301$ country-waves across 98 training countries). We evaluate both Out-of-Bag (`OOB`) predictions and strict 4-fold country-blocked cross-validation (`CV`) within the training set. Table 4, Figure 2, and Figure 3 report the Random Forest fit diagnostics and orthogonal residual support ($Z_{i,t} - \hat{z}(X_{i,t})$) across income tiers.

Table 4: Random Forest expected-treatment fit ($\hat{z}(X_{i,t}) = \mathbb{E}[Z_{i,t} \mid X_{i,t}]$) and orthogonal residual overlap diagnostics across World Bank income groups in the 80% Training Set ($N = 301$ country-waves, 98 countries).

Scaling & Income Tier	Waves (N)	Countries	SD(Z)	**RF OOB R^2**	**RF CV R^2**	OOB RMSE	Residual SD($Z - \hat{z}$)	Residual IQR	Support ≤ 2.5 SD (%)
Scaling A: All Training	**301**	**98**	**2.57**	**0.554**	**0.343**	**1.71**	**1.72**	**1.04**	**95.0%**
Scaling A: Low income	17	8	2.76	0.000	0.000	3.24	2.55	3.78	70.6%
Scaling A: Lower middle	48	20	5.13	0.638	0.429	3.06	3.06	3.92	85.4%
Scaling A: Upper middle	82	30	2.06	0.421	0.185	1.56	1.56	1.26	96.3%
Scaling A: High income	154	40	1.06	0.665	0.330	0.61	0.61	0.60	100.0%
Scaling B: All Training	**301**	**98**	**3.00**	**0.576**	**0.360**	**1.95**	**1.95**	**1.67**	**97.3%**
Scaling B: Low income	17	8	2.62	0.069	0.043	2.45	2.12	2.65	**100.0%**
Scaling B: Lower middle	48	20	5.68	0.641	0.417	3.36	3.40	3.47	85.4%
Scaling B: Upper middle	82	30	2.76	0.476	0.265	1.99	1.99	2.44	**98.8%**
Scaling B: High income	154	40	1.68	0.590	0.331	1.07	1.07	1.01	**100.0%**

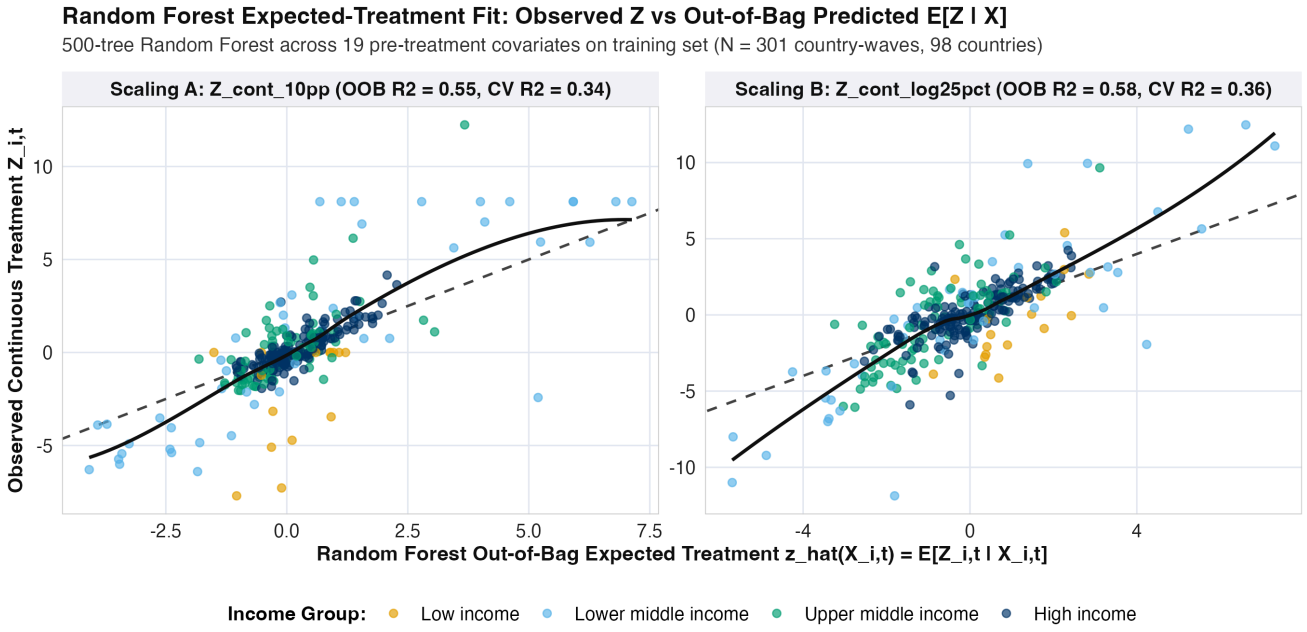


Figure 2: Observed continuous university funding Z versus Random Forest Out-of-Bag expected treatment E[Z | X] on the 80% Training Set (N = 301 country-waves across 98 countries) for Scaling A (left) and Scaling B (right).

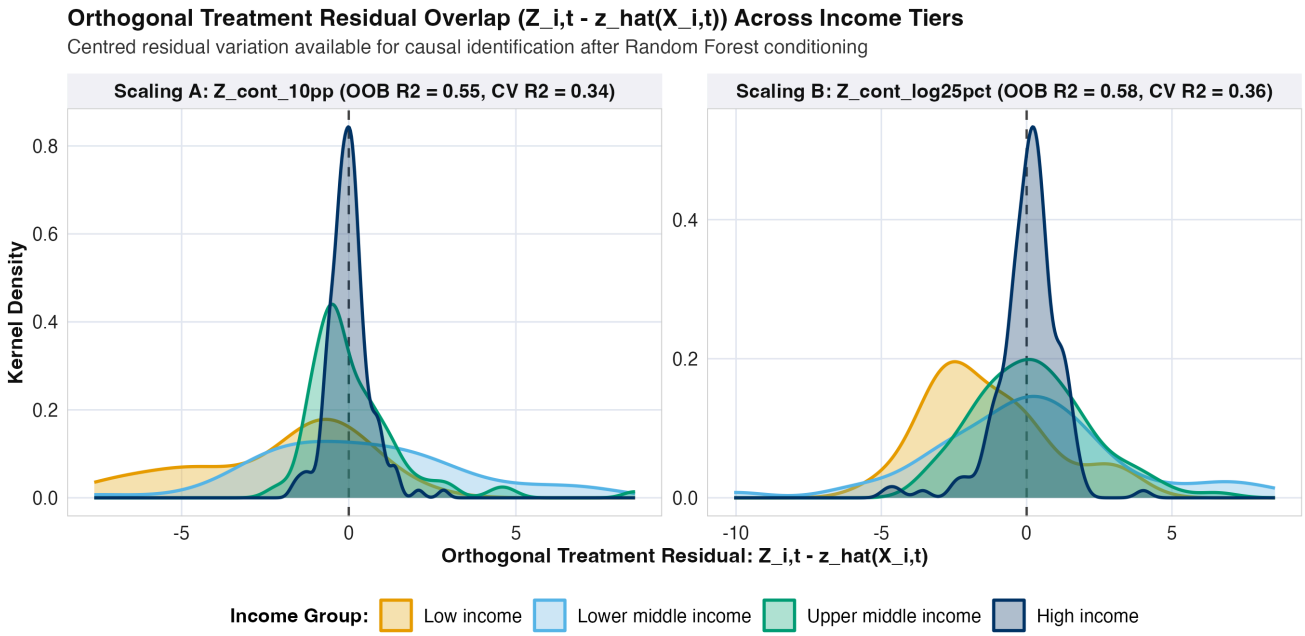


Figure 3: Kernel density distributions of the orthogonal treatment residuals Z - z_hat(X) across World Bank income groups for Scaling A (left) and Scaling B (right) on the 80% Training Set.

The Random Forest expected-treatment audit in Table 4, Figure 2, and Figure 3 establishes three foundational properties for our continuous-treatment causal identification strategy:

1. **Strong Predictive Fit Without Deterministic Collinearity (The Ideal Identification Regime):** Across the 98 training countries, the 500-tree Random Forest explains **55.4%** of the variance in `Z_cont_10pp` (`OOB R2 = 0.554`, 4-fold country-blocked `CV R2 = 0.343`) and **57.6%** of the variance in `Z_cont_log25pct` (`OOB R2 = 0.576`, country-blocked `CV R2 = 0.360`). Permutation importance shows that total public education budget share of GDP (`edu_spend_total_gdp_pct`), gross tertiary and secondary enrolment (`gross_tert_enrolment_pct`, `gross_seco_enrolment_pct`), baseline tax-to-GDP (`non_resource_tax_gdp_pct`), and lagged GDP growth (`lag_gdp_pc_growth_pct`) are the dominant predictors of $\hat{z}(X_{i,t})$. Because R^2 is substantial ($\approx 0.56 - 0.58$), omitting $\hat{z}(X_{i,t})$ would leave severe confounding; because $1 - R^2 \approx 0.42 - 0.45$ remains unexplained by baseline covariates, there is ample independent policy variation ($\text{SD}(Z_{i,t} - \hat{z}(X_{i,t})) = 1.72$ units under Scaling A and 1.95 units under Scaling B) to identify the causal slope $\tau(X_{i,t})$ without positivity violation.
2. **Superior Cross-Income-Tier Calibration and Residual Overlap Under Scaling B (`Z_cont_log25pct`):** Under Scaling A (`Z_cont_10pp`), the `150%` winsorisation ceiling clamps over half of Low income observations, resulting in `OOB R2 = 0.000` and only `70.6%` common support within ± 2.5 SD in Low income states, alongside a very narrow residual spread in High income states (IQR $= 0.60$). Under Scaling B (`Z_cont_log25pct`), the log-relative transformation achieves higher overall out-of-bag and country-blocked cross-validated fit (`OOB R2 = 0.576`, `CV R2 = 0.360`), raises overall common support to **97.3%** (`100.0%` in Low income, `98.8%` in Upper middle income, and `100.0%` in High income), and equalises the orthogonal residual distributions across income tiers (Figure 3).
3. **Why We Test Both Scalings in Phase 4:** Given that both scalings exhibit well-behaved, zero-centred orthogonal residuals $Z_{i,t} - \hat{z}(X_{i,t})$ across all four development tiers, Phase 4 will estimate the Multivariate BCF under both Scaling A and Scaling B and compare their posterior residuals, WAIC / log-predictive density, and held-out country prediction accuracy.

6 Missing Data Analysis on the Training Set

Table 5 audits missingness rates across the 80% Training Set (1,408 country-waves across 176 sovereign countries), within the complete +5-year training sample ($N = 301$), and across Low versus High income training economies.

Table 5: Missing-data rates across the 80% Training Set (1,408 country-waves) and within the $N = 301$ +5-year training sample.

Variable Name	Full Train Missing (%)	+5 y Train Sample (%)	Low Income Train (%)	High Income Train (%)
Per-Student Tertiary Ratio (% GDP/pc)	70.3%	**0.0%**	76.2%	65.2%
Continuous Z Scaling A (Z_cont_10pp)	70.3%	**0.0%**	76.2%	65.2%
Continuous Z Scaling B (Z_cont_log25pct)	70.3%	**0.0%**	76.2%	65.2%
Real 2021 PPP $ per Tertiary Student	70.8%	**0.0%**	76.2%	65.9%
Public Tertiary Budget (% of GDP)	54.8%	**6.3%**	60.7%	52.4%
Non-Resource Tax Revenue (% of GDP)	46.7%	**0.0%**	64.9%	43.3%
Direct Income & Profit Share (% Rev)	47.6%	**0.0%**	66.1%	44.2%
Goods & Services Tax Share (% Rev)	47.4%	**0.0%**	66.1%	44.0%
Log Real GDP per Person (PPP)	20.8%	**0.0%**	26.2%	26.6%
Lagged ($t-1$) GDP per Person Growth (%)	18.9%	**0.0%**	21.4%	22.6%
Gross Tertiary Enrolment (%)	35.5%	**11.0%**	31.5%	48.6%
Harmonised Disposable Gini Index	30.8%	**2.7%**	31.5%	38.2%
Trade Openness (% of GDP)	21.2%	**4.7%**	28.0%	24.5%
Total Unemployment Rate (%)	24.4%	**0.7%**	12.5%	36.6%

As documented in Table 5, treatment exposures and four-channel tax outcomes are **never imputed** (0.0% missingness in the estimation sample). Missingness across the historical 1985-2024 grid is higher in Low income states (64.9% to 76.2%) than in High income states (43.3% to 65.2%), consistent with a Missing at Random (MAR) mechanism governed by administrative capacity and income group, which are explicitly conditioned on in both $\hat{z}(X_{i,t})$ and $\mu(X_{i,t}, \hat{z}_{i,t})$. Within the training estimation sample, covariate missingness is minimal (0.0% for log GDP per person and lagged growth, 0.7% for unemployment, 2.7% for disposable Gini, 4.7% for trade openness, and 11.0% for tertiary enrolment) and is handled via training-fold median imputation paired with explicit binary missingness indicators (plus a complete-case sensitivity check R4 dropping all waves with any missing covariate).

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7 Temporal Autocorrelation, Stationarity, and Spatial Clustering

Table 6 compares within-country temporal autocorrelation (lag1_acf, lag2_acf), sovereign country and World Bank regional Intraclass Correlation Coefficients (country_icc, region_icc), and panel stationarity p-values across baseline fiscal levels ($Y_{i,t}$), continuous treatments ($Z_{i,t}$), and 5-year first-differenced fiscal changes ($\Delta Y_{i,t+1}$) on the 80% Training Set.

Table 6: Training-set temporal autocorrelation, sovereign country and regional Intraclass Correlation Coefficients (ICC), and panel stationarity diagnostics.

Series Name	Series Type	Lag-1 ACF (r_1)	Lag-2 ACF (r_2)	Country ICC	Region ICC	Panel ADF p-value
1. Non-Resource Tax / GDP (%)	Level (Y_t)	**0.941**	0.894	**0.883**	0.164	< 0.001
2. Direct Income Tax / GDP (%)	Level (Y_t)	**0.944**	0.914	**0.873**	0.144	< 0.001
3. Direct Income Tax Share (%)	Level (Y_t)	**0.952**	0.909	**0.859**	0.242	< 0.001
4. Goods & Services Share (%)	Level (Y_t)	**0.907**	0.802	**0.775**	0.188	< 0.001
5. Continuous Z (`Z_cont_10pp`)	Treatment (Z_t)	**0.877**	0.633	**0.768**	0.075	< 0.001
6. Continuous Z (`Z_cont_log25pct`)	Treatment (Z_t)	**0.887**	0.701	**0.790**	0.046	< 0.001
7. Δ Non-Resource Tax / GDP (pp)	Change (ΔY_{t+1})	**-0.072**	−0.103	**0.000**	0.000	< 0.001
8. Δ Direct Income Tax / GDP (pp)	Change (ΔY_{t+1})	**-0.054**	−0.155	**0.068**	0.000	< 0.001
9. Δ Direct Income Tax Share (pp)	Change (ΔY_{t+1})	**+0.018**	−0.050	**0.000**	0.012	< 0.001
10. Δ Goods & Services Share (pp)	Change (ΔY_{t+1})	**+0.022**	−0.149	**0.006**	0.018	< 0.001

Table 6 confirms why first-differenced fiscal changes ($\Delta Y_{i,t+h}$) and country-clustered inference are essential: raw fiscal levels exhibit near-unit-root persistence (`lag1_acf` of `0.907` to `0.952`; `country_icc` of `0.775` to `0.883`), whereas 5-year percentage-point changes are stationary (`lag1_acf` between `-0.072` and `+0.022`; `country_icc` between `0.000` and `0.068`). Moreover, because continuous university funding intensity $Z_{i,t}$ is highly persistent within sovereign states (`country_icc = 0.768` for Scaling A and `0.790` for Scaling B), our holdout validation splits entire sovereign countries (`country_fold`) rather than individual country-waves so that no country's historical trajectory leaks between training and test sets.

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8 Feature Correlation Structure and Variance Inflation Factors

Finally, Table 7 and Figure 4 audit collinearity among the 15 continuous pre-treatment baseline covariates $X_{i,t}$ in the 80% Training Set ($N = 301$) and their pairwise correlations with `Z_cont_10pp`, `Z_cont_log25pct`, and $+5$ -year fiscal changes.

Table 7: Training-set Variance Inflation Factors (VIFs) for the continuous pre-treatment baseline confounders and pairwise correlations with continuous university funding (Z_{10pp}, $Z_{log25\%}$) and +5-year fiscal outcomes ($N = 301$).

Baseline Covariate ($X_{i,t}$)	VIF	Corr. with Z_{10pp}	Corr. with $Z_{log25\%}$	Corr. with $\Delta\text{Tax/GDP}_{+5\text{ y}}$	Corr. with $\Delta\text{IncShare}_{+5\text{ y}}$
1. Log Real GDP per Person (PPP)	**7.06**	+0.046	+0.165	−0.138	−0.071
2. Baseline Life Expectancy (Years)	**5.65**	−0.242	−0.076	−0.065	+0.015
3. Gross Secondary Enrolment (%)	**4.19**	−0.031	+0.070	−0.112	−0.096
4. Gross Tertiary Enrolment (%)	**4.14**	−0.130	−0.049	−0.131	−0.126
5. Working-Age Population Share (%)	**3.22**	−0.170	−0.106	−0.053	−0.111
6. Baseline Non-Resource Tax / GDP (%)	**3.05**	**+0.263**	**+0.289**	**-0.181**	−0.064
7. Baseline Direct Income Tax Share (%)	2.04	+0.090	+0.177	−0.081	−0.140
8. Baseline Goods & Services Share (%)	2.02	−0.156	−0.132	+0.046	+0.128
9. Log Population Size	1.93	−0.149	−0.104	−0.091	−0.012
10. Baseline Disposable Gini Index	1.82	+0.166	+0.015	+0.136	+0.094
11. Total Education Spend (% GDP)	1.81	**+0.497**	**+0.525**	−0.018	−0.032
12. Trade Openness (% GDP)	1.63	+0.074	+0.111	−0.039	−0.026
13. Natural Resource Rents (% GDP)	1.54	+0.078	+0.045	+0.030	+0.005
14. Unemployment Rate (%)	1.43	+0.192	+0.123	+0.075	−0.011
15. Lagged ($t-1$) GDP Growth (%)	1.16	−0.246	−0.296	+0.069	+0.026

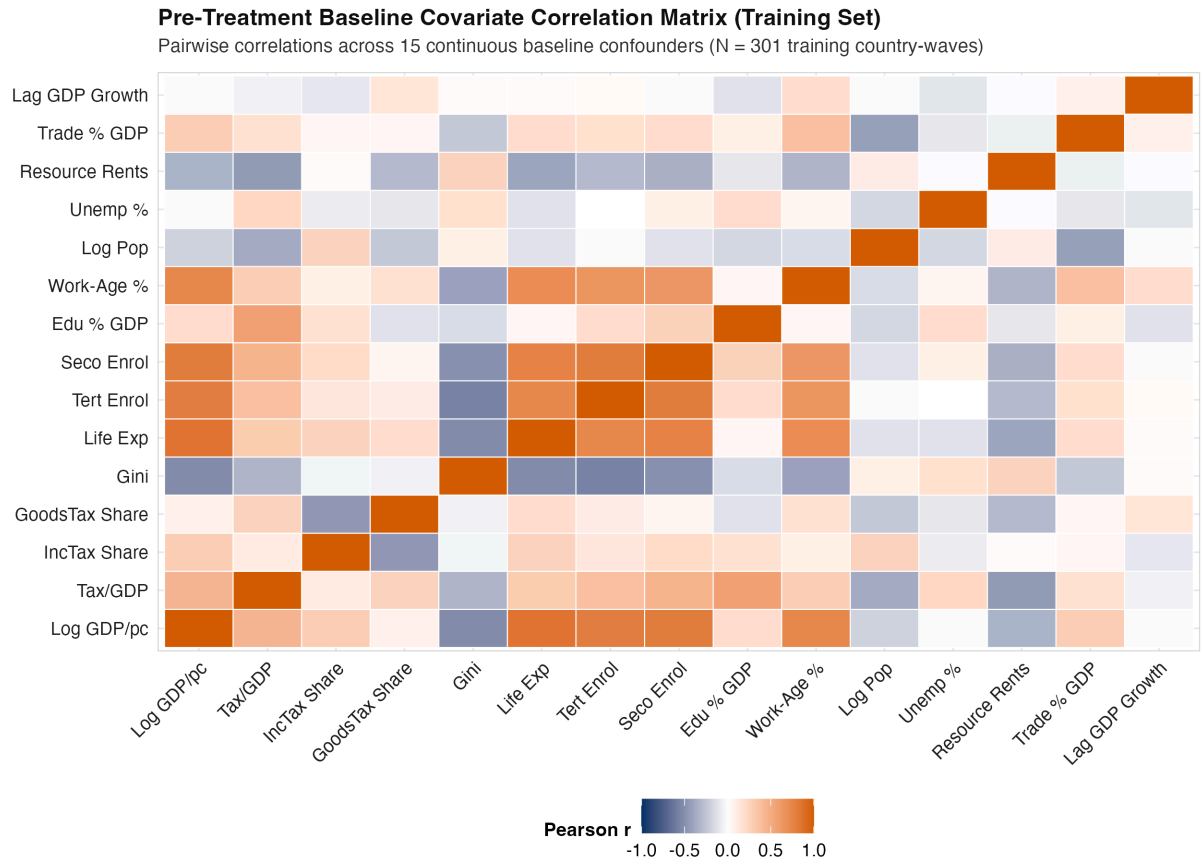


Figure 4: Training-set pairwise Pearson correlation matrix across the 15 continuous pre-treatment baseline covariates (N = 301 country-waves across 98 training countries).

All 15 continuous baseline covariates have Variance Inflation Factors comfortably below `10.0` (peaking at `7.06` for log GDP per person and `5.65` for life expectancy), and no predictor shows target leakage against forward fiscal changes ($\max|r| = 0.181$). Crucially, Table 7 highlights why covariate adjustment is indispensable: baseline non-resource tax-to-GDP correlates positively with continuous university funding ($r = +0.263$ for Scaling A and $+0.289$ for Scaling B) but negatively with subsequent tax-to-GDP changes ($r = -0.181$) due to fiscal mean-reversion, meaning that unadjusted regressions suffer from downward omitted-variable bias unless baseline fiscal capacity is conditioned out via $\mu(X_{i,t}, \hat{z}_{i,t})$.